MULTIVARIABLE CALCULUS • UNIT 2.01

C Multivariable Chain Rule, Directional Derivatives & Gradients

PRACTICE WORKSHEET

25 Structured Exam-Style Questions | Name: _____ | Date: _____ | Score: _____ / 25

Multivariable Chain Rule, Directional Derivatives & Gradients Practice Exercises

1 MCQ

Find $\nabla f(-5, 3)$ for $f(x, y) = 4x^2 + 3y^2$.

A (18, -40)

B (-20, 9)

C (-40, 18)

D (40, -18)

2 MCQ

Find $\nabla f(-6, 3)$ for $f(x, y) = 3x^2 + y^2$.

A (6, -36)

B (-18, 3)

C (-36, 6)

D (36, -6)

3 MCQ

Find $\nabla f(5, -1)$ for $f(x, y) = 3x^2 + 5y^2$.

A (-10, 30)

B (30, -10)

C (15, -5)

D (-30, 10)

4 MCQ

Find $\nabla f(-5, -5)$ for $f(x, y) = 9x^2 + 6y^2$.

A (-90, -60)

B (-60, -90)

C (-45, -30)

D (90, 60)

5 MCQ

Find $\nabla f(-2, 2)$ for $f(x, y) = 2x^2 + 7y^2$.

A (28, -8)

B (-4, 14)

C (8, -28)

D (-8, 28)

6 MCQ

Find $\nabla f(-2,-1)$ for $f(x,y) = 4x^2 + 8y^2$.

A (-8, -8)

B (16, 16)

C (-16, -16)

D (-16, -14)

7 MCQ

Find $\nabla f(-2,-4)$ for $f(x,y) = 2x^2 + 6y^2$.

A (-48, -8)

B (-8, -48)

C (-4, -24)

D (8, 48)

8 MCQ

Find $\nabla f(-1,-4)$ for $f(x,y) = 4x^2 + 8y^2$.

A (-64, -8)

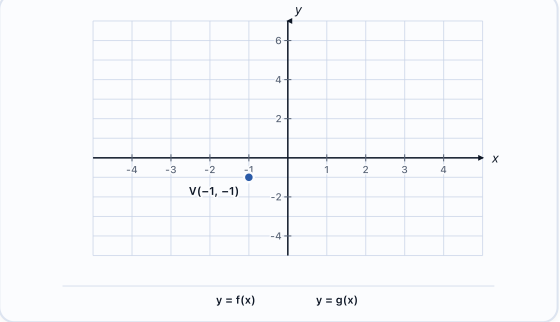
B (-4, -32)

C (-8, -64)

D (8, 64)

9 MCQ

Consider the function $z = f(x,y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.



A 26

B 10

C 13

D 52

10 MCQ

Consider the vector field $\mathbf{f}(x,y,z) = \langle 12x,\ 20y,\ 11z \rangle$ on R^3. Determine the scalar divergence $\nabla \cdot \mathbf{f}$.

A 23

B 43

C 47

D 55

11 MCQ

For $f(x, y) = 8x - 15y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 23
B 16
C 17
D 18

12 MCQ

For $f(x, y) = 4x - 3y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 7
B 4
C 6
D 5

13 MCQ

For $f(x, y) = 8x - 6y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 14
B 10
C 9
D 11

14 MCQ

For $f(x, y) = 24x - 7y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 31
B 24
C 26
D 25

15 MCQ

For $f(x, y) = -20x + 21y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 29
B 41
C 28
D 30

16 MCQ

For $f(x, y) = -15x + 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 23
B 16
C 18
D 17

17 MCQ

For $f(x, y) = -6x - 8y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 10
B 14
C 9
D 11

18 MCQ

For $f(x, y) = -3x - 4y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 6
B 7
C 5
D 4

19 MCQ

For $f(x, y) = 12x - 5y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 13
B 12
C 14
D 17

20 MCQ

For $f(x, y) = -9x + 40y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 40
B 41
C 42
D 49

21 MCQ

For $f(x, y) = -4x + 3y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 7
B 5
C 4
D 6

22 MCQ

For $f(x, y) = -3x + 4y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 5
B 7
C 4
D 6

23 MCQ

For $f(x, y) = 12x + 35y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 47

B 36

C 38

D 37

24 MCQ

For $f(x, y) = -9x - 40y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 41

B 40

C 42

D 49

25 MCQ

For $f(x, y) = -12x - 35y$, find the maximum possible value of the directional derivative $D_{\mathbf{u}}f$ at any point, over all unit vectors $\mathbf{u}$.

A 47

B 36

C 37

D 38